

A. Basic Data Cleaning — 15 Marks

1. Rename unclear columns. (2 Marks)

FileHomeTransformAdd ColumnViewToolsHelp

Group ByUse First Row as HeadersCount Rows

TransposeReverse RowsCount Rows

Data Type: TextDetect Data TypeRename

Any Column

StatisticsStandardScientific

TrigonometryRoundingInformation

DateTimeDuration

Run R scriptRun Python script

Queries [3]

OrdersPeopleReturns

This preview may be up to 7 days old. Refresh

Table.TransformColumnTypes(#"Promoted Headers",{{"Person", type

Person NameRegion

Valid100%Valid100%

Error0%Error0%

Empty0%Empty0%

24 distinct, 24 unique24 distinct, 24 unique

1	Mari��ne Rousseau	Caribbean
2	Andile Ihejirika	Central Africa
3	Nicodemo Bautista	Central America
4	Cansu Peynirci	Central Asia
5	Lon Bonher	Central US
6	Wasswa Ahmed	Eastern Africa
7	Hadia Bousaid	Eastern Asia
8	Lynne Marchand	Eastern Canada
9	Oxana Lagunov	Eastern Europe
10	Dolores Davis	Eastern US
11	Lindiwe Afolayan	North Africa
12	Miina Nylund	Northern Europe
13	Kauri Anaru	Oceania
14	Vasco Magalh��es	South America
15	Preecha Metharom	Southeastern Asia
16	Nora Cui��er	Southern Africa

2 COLUMNS, 24 ROWSColumn profiling based on top 1000 rows

Query Settings

PROPERTIESNamePeopleAll Properties

APPLIED STEPS

SourceNavigationChanged TypePromoted Headers

Changed Type1

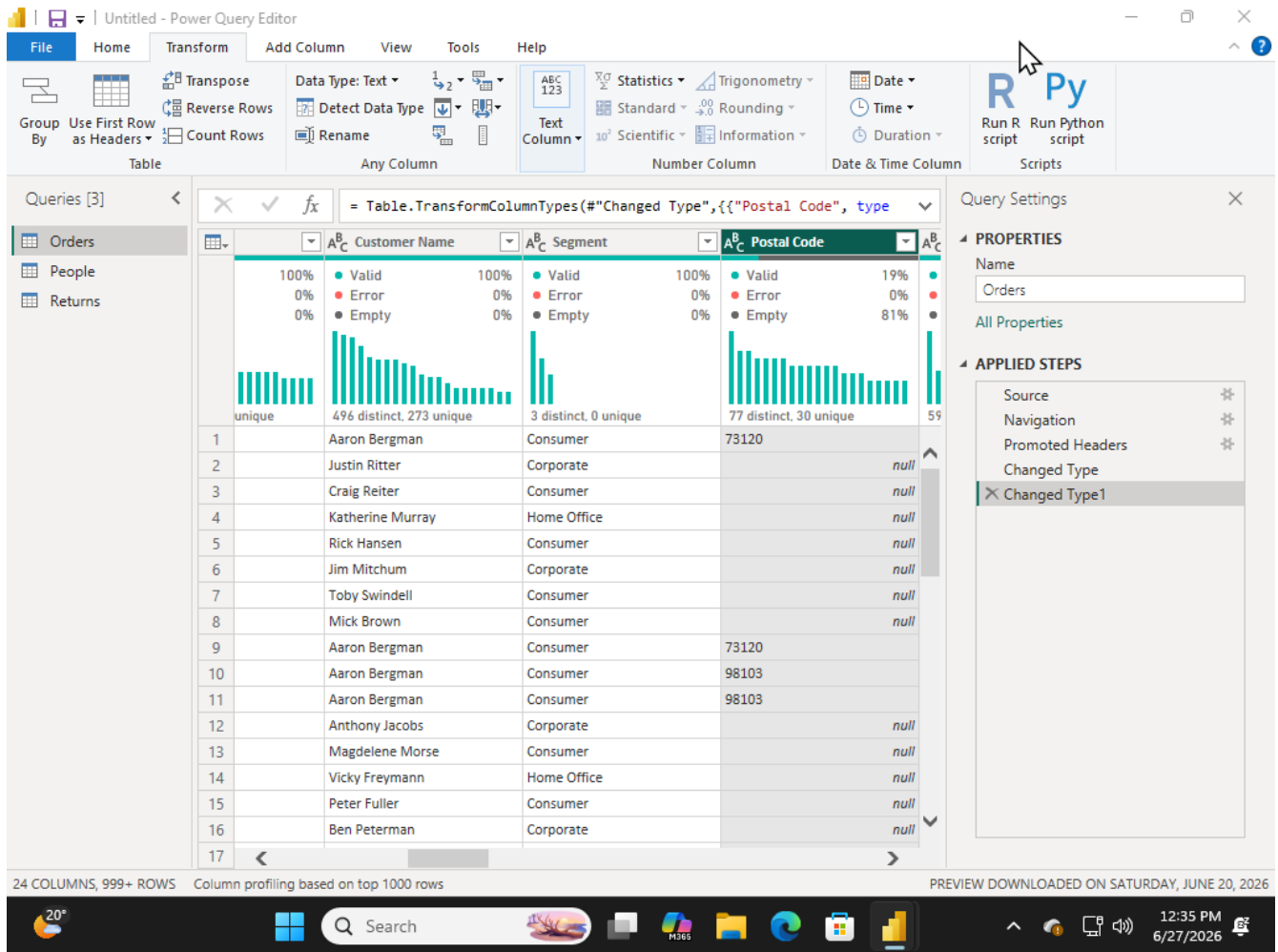
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Search

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2. Change data types correctly. (2 Marks)



3. Remove duplicate records. (2 Marks)

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Data Type: Any Detect Data Type Rename

Any Column

Text Column

Statistics Standard Scientific

Trigonometry Rounding Information

Date Time Duration

Date & Time Column

Run R script Run Python script

Scripts

Queries [3]

Orders

People

Returns

fx = Table.Distinct("#Changed Type1")

A^B_C City A^B_C State A^B_C Country A^B_C Region

Valid 100% Error 0% Empty 0%

592 distinct, 435 unique

Valid 100% Error 0% Empty 0%

341 distinct, 185 unique

Valid 100% Error 0% Empty 0%

93 distinct, 35 unique

Valid 100% Error 0% Empty 0%

22 distinct

1 Oklahoma City Oklahoma United States Central

2 Wollongong New South Wales Australia Oceania

3 Brisbane Queensland Australia Oceania

4 Berlin Germany Western Europe

5 Dakar Senegal Western Africa

6 Sydney New South Wales Australia Oceania

7 Porirua Wellington New Zealand Oceania

8 Hamilton Waikato New Zealand Oceania

9 Oklahoma City Oklahoma United States Central

10 Seattle Washington United States Western North America

11 Seattle Washington United States Western North America

12 Kabul Kabul Afghanistan Southern Asia

13 Jizan Jizan Saudi Arabia Western Asia

14 Toledo Parana Brazil South America

15 Mudanjiang Heilongjiang China Eastern Asia

16 Paris Ile-de-France France Western Europe

17

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Changed Type1

Removed Duplicates

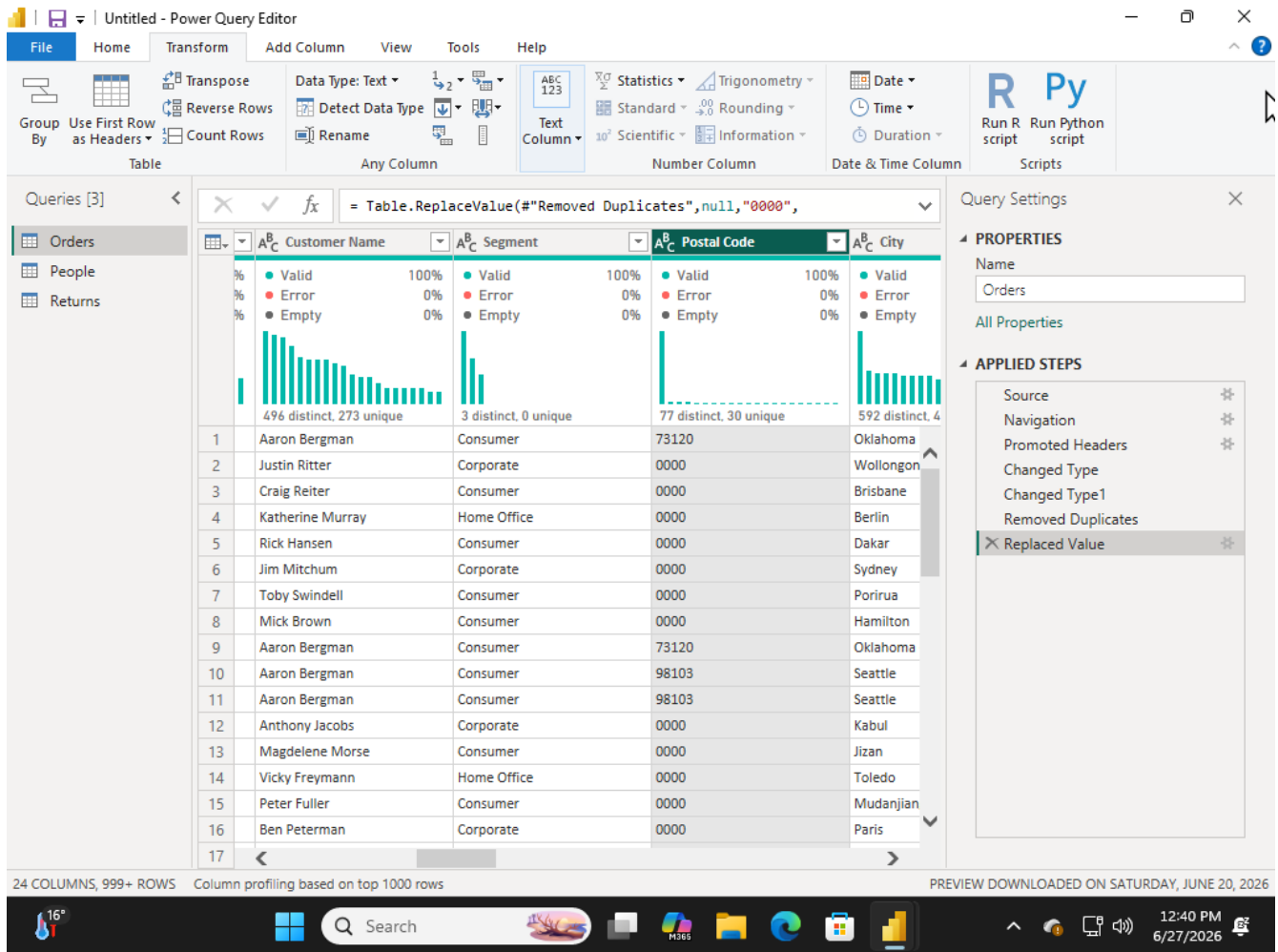
24 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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4. Remove blank rows. (2 Marks)



5. Trim and clean text columns. (2 Marks)

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Group By Use First Row as Headers Count Rows

Table

Any Column

Text Column

Statistics Standard Scientific

Trigonometry Rounding Information

Date Time Duration

Run R script Run Python script

Scripts

Queries [3]

Orders

People

Returns

Person Name

Region

Valid 100%

Error 0%

Empty 0%

24 distinct, 24 unique

24 distinct, 24 unique

1 Marilène Rousseau Caribbean

2 Andile Ihejirika Central Africa

3 Nicodemo Bautista Central America

4 Cansu Peynirci Central Asia

5 Lon Bonher Central US

6 Wasswa Ahmed Eastern Africa

7 Hadia Bousaid Eastern Asia

8 Lynne Marchand Eastern Canada

9 Oxana Lagunov Eastern Europe

10 Dolores Davis Eastern US

11 Lindiwe Afolayan North Africa

12 Miina Nylund Northern Europe

13 Kauri Anaru Oceania

14 Vasco Magalhães South America

15 Preecha Metharom Southeastern Asia

16 Nora Cuijper Southern Africa

17 Chandrakant Chaudhri Southern Asia

2 COLUMNS, 24 ROWS Column profiling based on top 1000 rows

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Search

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Query Settings

PROPERTIES

Name

People

All Properties

APPLIED STEPS

Source

Navigation

Changed Type

Promoted Headers

Changed Type1

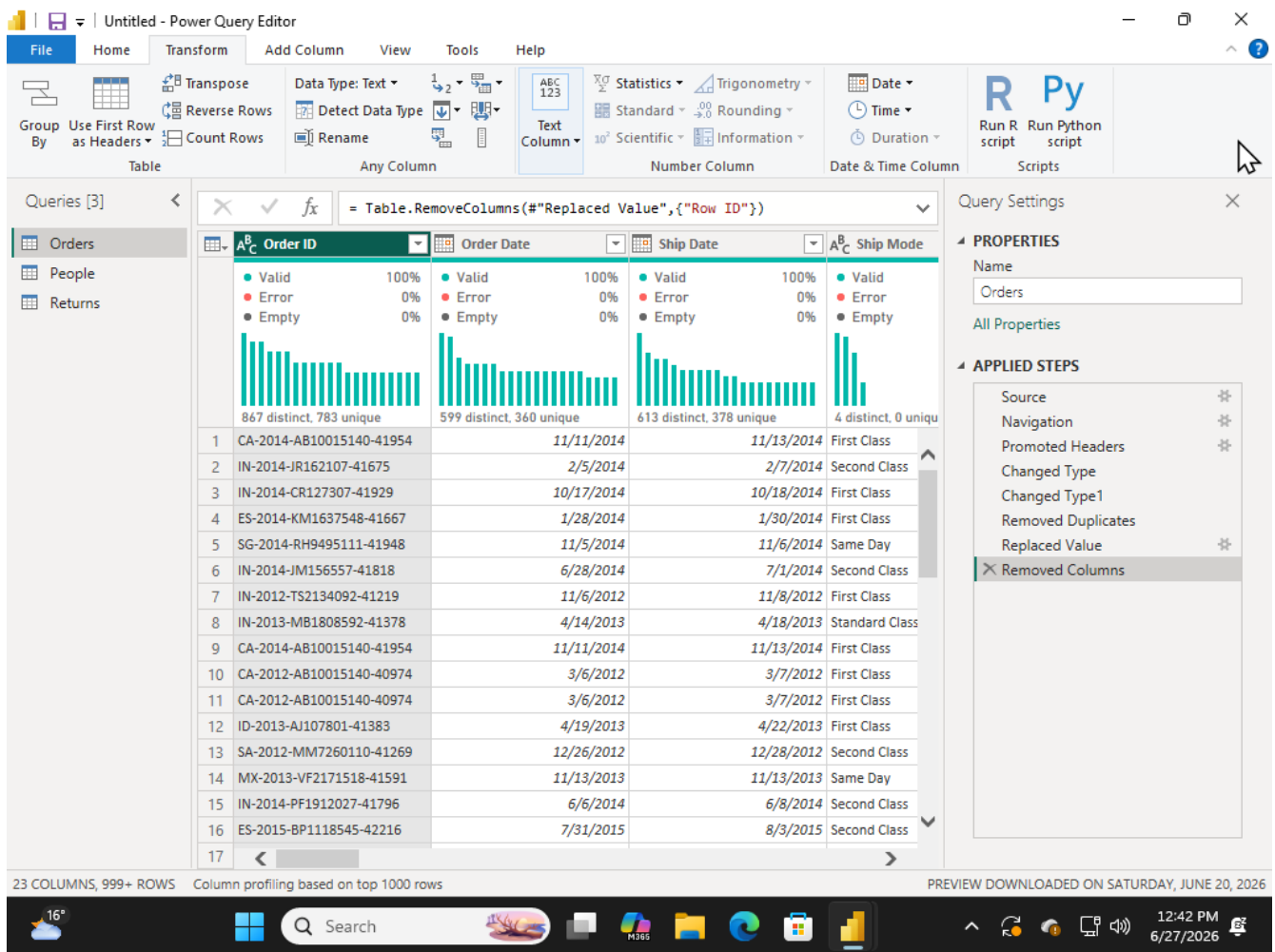
Renamed Columns

Trimmed Text

6. Replace inconsistent values. (2 Marks)

No inconsistent values

7. Remove unnecessary columns. (3 Marks)



1. Split one column into multiple columns. (2 Marks)

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Layout Data Preview Columns Parameters Advanced Dependencies

Queries [6] People Returns Orders Orders_dup Top Products Category_Region_S...

Table.TransformColumnTypes(#"Promoted Headers",{"Returned", type text},

Split Column by Number of Characters

Specify the number of characters used to split the text column.

Number of characters:

Split

☐ Once, as far left as possible

☐ Once, as far right as possible

☒ Repeatedly

Advanced options

OK Cancel

Column statistics

Count	1000
Error	0
Empty	0
Distinct	24
Unique	1

Value distribution

Western Europe	
Central America	
Oceania	
Western US	
Eastern US	

Query Settings

PROPERTIES

Name: Returns

All Properties

APPLIED STEPS

Source

Navigation

Changed Type

Promoted Headers

Changed Type1

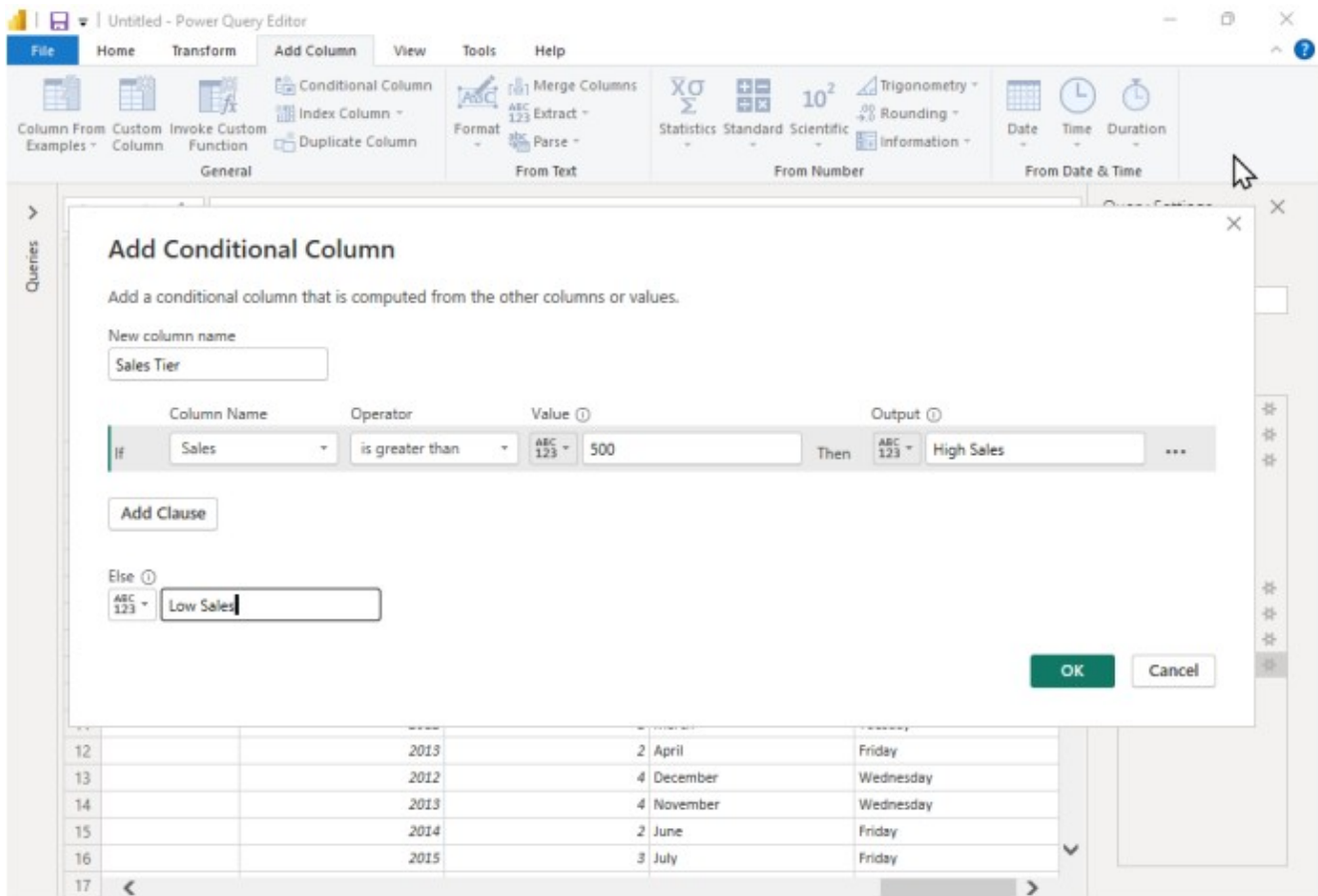
3 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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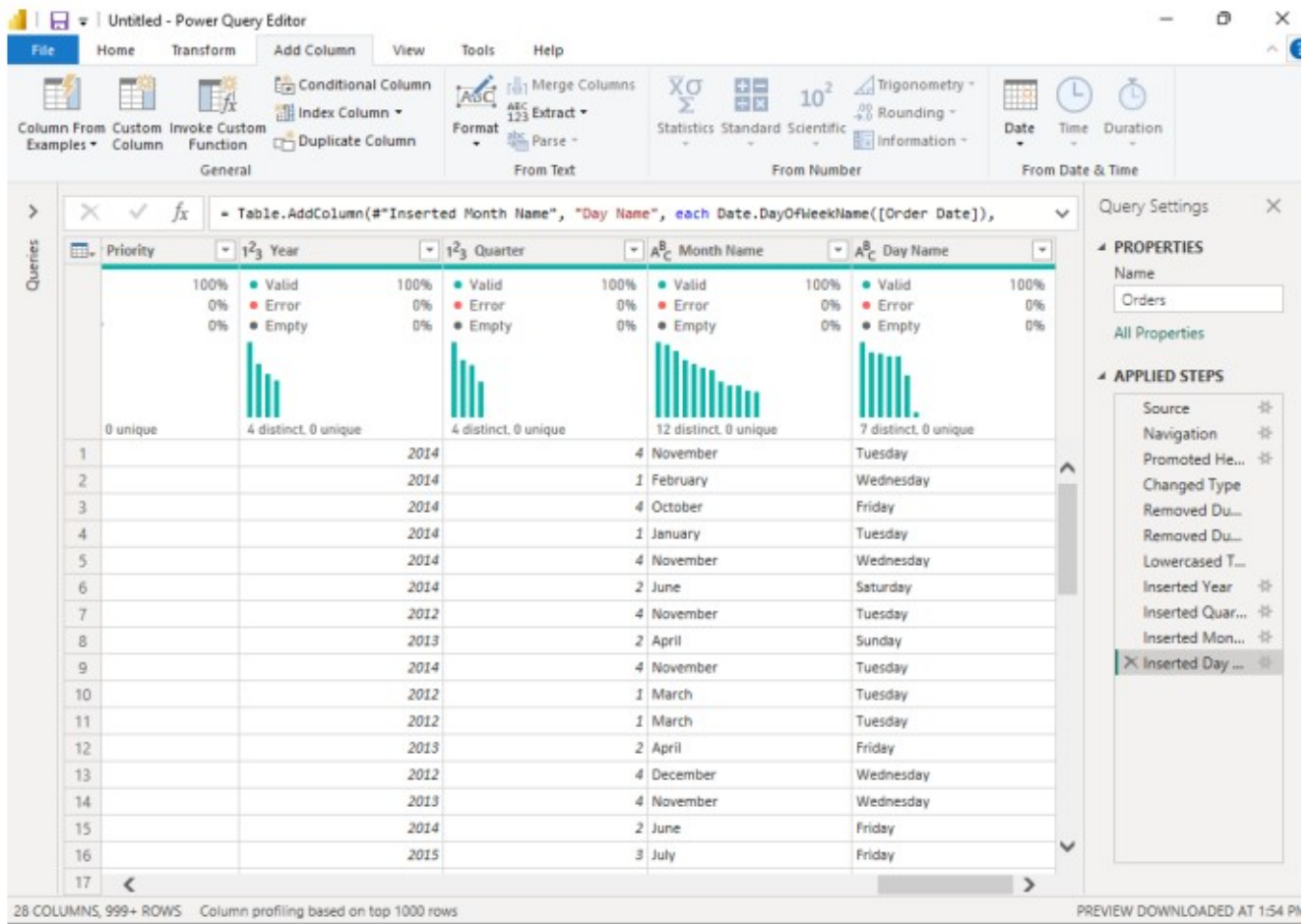
2. Merge two or more columns. (2 Marks)

3. Create a custom column. (3 Marks)

4. Create a conditional column. (3 Marks)



5. Extract year, month, quarter, and day from a date column. (4 Marks)



6. Filter rows using at least two conditions. (2 Marks)

Power Query Editor interface showing a table with columns: Month Name, Day Name, Sales Tier, and Customer Value Tier. The ribbon includes options like Add Column, View, Tools, and Help. The right sidebar shows Query Settings and Applied Steps.

Table Data:

Month Name	Day Name	Sales Tier	Customer Value Tier
1 November	Tuesday	Low Sales	Silver
2 ebruary	Wednesday	High Sales	Bronze
3 ctober	Friday	High Sales	Gold
4 inuary	Tuesday	High Sales	Bronze
5 ovember	Wednesday	High Sales	Gold
6 ne	Saturday	High Sales	Gold
7 ovember	Tuesday	High Sales	Gold
8 pril	Sunday	High Sales	Gold
9 ovember	Tuesday	Low Sales	Silver
10 arch	Tuesday	Low Sales	Bronze
11 arch	Tuesday	Low Sales	Bronze
12 pril	Friday	High Sales	Gold
13 eember	Wednesday	High Sales	Gold
14 ovember	Wednesday	High Sales	Gold
15 ne	Friday	High Sales	Gold
16 uly	Friday	High Sales	Silver

Applied Steps:

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Removed Duplicates
- Lowercase Text
- Inserted Year
- Inserted Quarter
- Inserted Month
- Inserted Day
- Added Conditional Column
- Added Conditional Column

7. Sort data meaningfully. (2 Marks)

Untitled - Power Query Editor

FileHomeTransformAdd ColumnViewToolsHelp

Group ByUse First Row as HeadersCount Rows

Table

Any Column

Number Column

Date & Time Column

Scripts

TransposeReverse Rows

Data Type: DateDetect Data TypeRename

12ABC123Text Column

StatisticsStandardScientific

TrigonometryRoundingInformation

DateTimeDuration

Run R scriptRun Python script

Queries [3]

OrdersPeopleReturns

This preview may be up to 7 days old. Refresh

fx

= Table.RemoveColumns(#"Replaced Value",{"Row ID"})

ABOrder ID

Order Date

Ship Date

ABShip Mode

Sort AscendingSort DescendingClear SortClear FilterRemove EmptyDate Filters

Search

(Select All)

1/8/2012

1/10/2012

1/21/2012

1/24/2012

2/10/2012

2/14/2012

2/15/2012

2/16/2012

2/19/2012

2/23/2012

List may be incomplete. Load more

OKCancel

Valid 100%Error 0%Empty 0%

613 distinct, 378 unique

4 distinct, 0 unique

11/13/2014First Class

2/7/2014Second Class

10/18/2014First Class

1/30/2014First Class

11/6/2014Same Day

7/1/2014Second Class

11/8/2012First Class

on

9/18/2015

11/7/2013

12/18/2014

4/9/2015

12/16/2012

8/3/2013

9/2/2014

3/4/2012

11/7/2015

12/26/2013

12/7/2013

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Changed Type1

Removed Duplicates

Replaced Value

Removed Columns

23 COLUMNS, 999+ ROWS

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Search

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8. Add an index column. (2 Marks)

The screenshot shows the Power Query Editor interface. The 'Add Column' tab is active, and the 'Index Column' option is selected. The main area displays a table with the following columns: '1.2 Shipping Cost', 'A8C Order Priority', and '123 Index'. The 'Index' column is highlighted in green. The right sidebar shows the 'Query Settings' for 'Orders', including 'Properties' and 'Applied Steps'.

Index	1.2 Shipping Cost	A8C Order Priority
1	62.1544	High
2	-288.765	Critical
3	919.971	Medium
4	-96.54	Medium
5	311.52	Critical
6	763.275	Critical
7	564.84	Critical
8	996.48	High
9	54.7136	High
10	5.4801	High
11	4.6644	High
12	647.55	High
13	1151.4	Critical
14	622.02	Critical
15	1036.08	Critical
16	186.948	Critical

Complete any **five** of the following tasks.

Each task carries **5 marks**.

1. Merge queries using a common key.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

New Source Recent Sources Enter Data New Query

Queries [3]

- Orders
- People
- Returns

Merge

Select a table and matching columns to create a merged table.

Orders

Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment
CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class	AB-100151402	Aaron Bergman	Consumer
IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class	JR-162107	Justin Ritter	Corporate
IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class	CR-127307	Craig Reiter	Consumer
ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class	KM-1637548	Katherine Murray	Home Office

Returns

Returned	Order ID	Region
Yes	CA-2012-SA20830140-41210	Central US
Yes	IN-2012-PB19210127-41259	Eastern Asia
Yes	CA-2012-SC20095140-41174	Central US
Yes	IN-2015-JH158207-42140	Oceania
Yes	IN-2014-LC168857-41747	Oceania

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

Fuzzy matching options

Estimating matches based on data previews

OK Cancel

Row as Headers
Values
Combine

Headers
type
type1
Duplicates
value
Columns
ex

24 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED ON SATURDAY, JUNE 20, 2026

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2. Use Group By with multiple aggregations.

FileHomeTransformAdd ColumnViewToolsHelp

New sourceRecent SourcesEnter Data

Data source settingsManage ParametersExport query resultsRefresh Preview

PropertiesAdvanced EditorManage

Manage ColumnsReduce RowsSort

Split ColumnGroup ByData Type: TextUse First Row as HeadersReplace Values

Combine

Queries [4]

OrdersPeopleReturnsCategory_Performa...

fx= Orders

Order IDOrder DateShip DateShip Mode

Group By

Specify the columns to group by and one or more outputs.

BasicAdvanced

Category

Add grouping

New column nameOperationColumn

Total_salesSumSales

Total_profitSumProfit

Add aggregation

OKCancel

Query SettingsPROPERTIES

25 COLUMNS, 999+ ROWSColumn profiling based on top 1000 rowsPREVIEW DOWNLOADED AT 12:59 PM

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Search

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3. Extract text before or after a delimiter.

The screenshot shows the Microsoft Power Query Editor interface. The ribbon at the top includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The 'Add Column' tab is active, showing options like Conditional Column, Index Column, Duplicate Column, Merge Columns, Extract, Parse, Format, From Text, From Number, and From Date & Time. The 'Queries' list on the left shows 'Orders', 'People', 'Returns', and 'Category_Performa...'. The main area displays a table with columns 'Index', 'Returns', and 'Text Before Delimiter'. The 'Text Before Delimiter' column contains values like 'AB', 'JR', 'CR', 'KM', 'RH', 'JM', 'TS', and 'MB'. Below the table, there are 'Column statistics' and 'Value distribution' charts. The 'Column statistics' chart shows counts for 'Count', 'Error', 'Empty', 'Distinct', and 'Unique'. The 'Value distribution' chart shows the frequency of values in the 'Text Before Delimiter' column. The right-hand pane shows 'Query Settings' for the 'Orders' query, including 'Properties' and 'Applied Steps'. The 'Applied Steps' list includes 'Source', 'Navigation', 'Promoted Headers', 'Changed Type', 'Changed Type1', 'Removed Duplicates', 'Replaced Value', 'Removed Columns', 'Added Index', 'Merged Queries', and 'Inserted Text Before Delimiter'.

26 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 12:59 PM

4. Create a summarized query using Reference Query.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

New Query Recent Sources Enter Data Data source settings Manage Parameters Export query results Refresh Preview Properties Advanced Editor Manage Columns Reduce Rows Sort Split Column Group By Use First Row as Headers Replace Values Combine

Queries [4]

- Orders
- People
- Returns
- Category_Performa...

fx = Orders

	Order ID	Order Date	Ship Date	Ship Mode
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
	867 distinct, 783 unique	599 distinct, 360 unique	613 distinct, 378 unique	4 distinct, 0 unique
1	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class
2	IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class
3	IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class
4	ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class
5	SG-2014-RH9495111-41948	11/5/2014	11/6/2014	Same Day
6	IN-2014-JM156557-41818	6/28/2014	7/1/2014	Second Class
7	IN-2012-TS2134092-41219	11/6/2012	11/8/2012	First Class
8	IN-2013-MB1808592-41378	4/14/2013	4/18/2013	Standard Class
9	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class
10	CA-2012-AB10015140-40974	3/6/2012	3/7/2012	First Class
11	CA-2012-AB10015140-40974	3/6/2012	3/7/2012	First Class
12	ID-2013-AJ107801-41383	4/19/2013	4/22/2013	First Class
13	SA-2012-MM7260110-41269	12/26/2012	12/28/2012	Second Class
14	MX-2013-VF2171518-41591	11/13/2013	11/13/2013	Same Day
15	IN-2014-PF1912027-41796	6/6/2014	6/8/2014	Second Class
16	ES-2015-BP1118545-42216	7/31/2015	8/3/2015	Second Class
17				

25 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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16° Search

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5. Use Column Profiling to identify data quality issues.

FileHomeTransformAdd ColumnViewToolsHelp

Query Settings

Layout

Formula Bar

Monospaced

Column distribution

Always allow

Show whitespace

Column profile

Advanced Editor

Column quality

Data Preview

Columns

Parameters

Advanced

Dependencies

Queries [4]

Orders

People

Returns

Category_Performa...

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

100%

0%

0%

Valid

Error

Empty

1000 distinct, 1000 unique

229 distinct, 59 unique

	Index	Returns	Text Before Delimiter
1		1 Table	AB
2		2 Table	JR
3		3 Table	CR
4		4 Table	KM
5		5 Table	RH
6		6 Table	JM
7		7 Table	TS
8		8 Table	MB
9			

Column statistics

Count

1000

Error

0

Empty

0

Distinct

229

Unique

59

Value distribution

AB

AH

AS

CS

CM

Query Settings

PROPERTIES

Name

Orders

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Changed Type1

Removed Duplicates

Replaced Value

Removed Columns

Added Index

Merged Queries

Inserted Text Before Delimiter

26 COLUMNS, 999+ ROWS

Column profiling based on top 1000 rows

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16°

Search

1:07 PM

6/27/2026